MORPHOLOGIC CHANGES IN RABBITS FOLLOWING THE INTRAVENOUS ADMINISTRATION OF MENINGOCOCCAL TOXIN

III. The Effects Produced by Endotoxin in Association with Certain High Molecular Weight Acidic Polymers *

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Previous papers[1,2] have dealt with the lesions occurring in rabbits following the intravenous administration of meningococcal endotoxin. The present paper reports the results of experiments designed to study the effects produced by meningococcal toxin when given in conjunction with certain high molecular weight acidic polymers, specifically sodium polyanetholsulfonate, the sodium salt of polyvinyl alcohol polysulfonic acid ester (PVAS), and dextran sulfate. Interest in these compounds was stimulated by the reports of Hausman and Dreyfus[3] and Walton[4] concerning the effects produced by these substances in animals and by the studies of Inderbitzin[5,6] concerning the experimental production of lupus erythematosus (LE) cells.

Sodium polyanetholsulfonate is a high molecular weight polymer of sodium anetholsulfonate manufactured by Hoffmann-La Roche, Inc., under the trade name of Liquoid-Roche. It has been used as an adjuvant in blood cultures[7] and as an anticoagulant *in vitro*,[8] but its exact mechanism of action is not known. Evidence indicates that at least part of its activity is due to a combination with fibrinogen.[9] Demole and Reinert[10] reported the development of widespread hemorrhages and a type of "glomerulonephritis" in rabbits given intravenous injections of Liquoid. Hausman and Dreyfus[3] described the occurrence of "intracapillary precipitates" in the renal glomeruli of rabbits given a single intravenous injection of a large amount of Liquoid, and similar precipitates with associated tubular necrosis were observed in rabbits given multiple injections of the material. The capillary lesions were described as being similar to the "wire loops" of disseminated lupus erythematosus and consisted of a homogeneous, acidophilic material lining the capillary basement membranes. The material was

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purple-red when stained by the periodic acid-Schiff method and purple with Weigert's fibrin stain. Similar material was noted in the pulmonary capillaries and in the liver sinusoids. These writers also described the occurrence of petechial hemorrhages in the thymus, heart, and lungs. In all but one animal, bilateral adrenal hemorrhage was observed.

Inderbitzin,[5] in 1952, reported the production of cells resembling LE cells by the addition of Liquoid to suspensions of normal leukocytes. In a later paper,[6] however, he reported that the inclusions in these cells appeared to represent phagocytosed platelets and nuclear débris rather than the amorphous material associated with true LE cells. In addition, he described the production of cells identical with those occurring in lupus erythematosus by the use of the sodium salt of the ester (PVAS) of polyvinyl alcohol and polysulfonic acid.

Little has been written concerning the biologic properties of PVAS. Patat and Vogler,[11] in 1952, described the preparation of a series of such substances and showed that they had anticoagulant properties somewhat similar to heparin. They also noted that the anticoagulant activity and the toxicity increased with increasing degrees of polymerization and sulfur content.

Reports by Ricketts and Walton[12] and Walton[4] concerning the biologic properties of certain dextran sulfate preparations indicated that these substances might have an action similar to that of Liquoid and PVAS. Walton[4] described the occurrence of "fibrin emboli" in the liver, spleen, lungs, and kidneys of animals given intravenous injections of dextran sulfates of varying molecular weights. These "emboli" were metachromatic when stained with toluidine blue, and similar metachromatic material was noted in the Kupffer cells of the liver and the reticulo-endothelial cells of the spleen. He reported that the toxicity of the dextran sulfates varied with the molecular weight; those of higher weights forming insoluble complexes with fibrinogen and producing agglutination of erythrocytes and platelets to form conglutination thrombi.

The renal lesion described by Hausman and Dreyfus[3] and Walton[4] seemed strikingly similar to that produced by two appropriately spaced intravenous injections of endotoxin (the generalized Shwartzman phenomenon). As reported previously,[2] the formation of fibrinoid is a characteristic feature of this phenomenon. It therefore seemed of interest to determine what changes might occur if endotoxin were given in conjunction with sodium polyanetholsulfonate, and the major portion of this report deals with those changes.

The results of these experiments then prompted us to investigate the changes occurring in animals given endotoxin in association with the sodium salt of PVAS, and with dextran sulfate. A more detailed account of these changes is the subject of another report.[13]

Materials and Methods

Meningococcal endotoxin, prepared by a method described previously,[14] was used throughout the experiments. It was diluted in sterile isotonic saline solution and injected in the marginal ear vein in a volume of 2 cc.

Liquoid and PVAS were obtained from Hoffmann-La Roche, Inc. The molecular weight of Liquoid is not known, but that of PVAS is reported to be approximately 40,000 and it has a sulfur content of approximately 18 per cent.[15] Dextran sulfate was prepared by the method of Ricketts and Walton.[12] These materials were dissolved in sterile, pyrogen-free, isotonic saline solution, passed through a Seitz filter, and injected in various doses in the marginal ear vein. Cultures of the filtered material were consistently sterile. In the experiments, 174 hybrid albino rabbits of mixed sexes were used. They were fed Purina rabbit pellets and had free access to water. In view of the previously described differences in reactivity of rabbits of different ages,[1] two sizes of animals were used. The "small" rabbits weighed between 1 and 1.5 kg., and the "large" rabbits weighed between 4 and 4.5 kg. Details concerning the numbers of animals injected with the materials under study are given in the tables.

The animals died or were killed at intervals ranging from 30 minutes to 96 hours following the last injection. Routine post-mortem examinations were performed and the tissues were fixed in 10 per cent neutral formalin. The hearts were sectioned so as to reveal the interventricular septum and the valves, and the brains were cut so as to include the choroid plexus. Hematoxylin and eosin were used routinely and selected sections from various organs were stained by one or more of the following methods: periodic acid-Schiff's, toluidine blue, phosphotungstic acid-hematoxylin, crystal violet, and van Gieson's stain.

Results

Lethal Effect

Liquoid Alone. In a dose of 4 mg., Liquoid caused no deaths, but with increasing amounts a corresponding increase in lethal effect was noted. With 32 mg., a majority of the small animals died within 12 hours following the injection. In the large rabbits, 16 mg. produced

death in one of 5 animals, and 32 mg. was lethal for 2 of 5 animals (Table I).

TABLE I
The Lethal Effect of Liquoid Alone

Group	Amount of Liquoid	Number of rabbits	Number dying in less than 12 hours
	mg.		
Small rabbits	4	12	0
	8	8	2
	16	10	3
	32	14	9
Large rabbits	16	5	1
	32	5	2

Endotoxin and Liquoid. The enhancement of the lethal effect of Liquoid by meningococcal toxin is shown in Table II. Seventy per cent of the small rabbits given 8 mg. of Liquoid 12 hours after an injection of endotoxin and 100 per cent of those given 8 mg. of the

TABLE II
The Lethal Effect of Liquoid in Conjunction with Meningococcal Toxin

Group	Dilution of toxin	Dose of Liquoid	Interval between injections	Number of rabbits	Number dying in less than 12 hours
		mg.	*hrs.*		
Small rabbits	1:80	4	18	3	0
			24	12	0
		8	6	8	8
			12	7	5
			18	22	2
			24	9	1
			30	3	0
	1:1000	8	Simultaneous	4	1
Large rabbits	1:1000	16	Simultaneous	5	4
		32	Simultaneous	5	4

material 6 hours after endotoxin died within 12 hours. A similar effect was noted in the large animals which were given minute amounts of endotoxin simultaneously with Liquoid; 80 per cent of these died within 12 hours after the injection. Further data concerning the minimum effective doses of endotoxin are contained in another report.[13]

Morphologic Changes

The structural changes following the administration of Liquoid and Liquoid in conjunction with endotoxin are summarized in Tables III and IV.

TABLE III

Structural Changes Following the Injection of Liquoid Alone

Group	Dose of Liquoid	Number of rabbits	Renal lesion	Cardiac lesions	Cardiac fibrinoid	Splenic fibrinoid	Liver thrombi and necrosis	Pulmonary thrombi and hemorrhage	Adrenal hemorrhage
	mg.								
	4	12	0	4	0	0	0	0	0
Small	8	8	1	4	1	0	0	4	0
rabbits	16	10	8	4	3	2	4	10	1
	32	14	9	3	0	0	2	14	1
Large	16	5	0	0	0	0	0	2	0
rabbits	32	5	4	0	1	3	1	5	4

TABLE IV

Structural Changes Following the Injection of Liquoid in Conjunction with Meningococcal Toxin

Group	Dilution of toxin	Dose of Liquoid	Interval between injections	Number of rabbits	Renal lesion	Cardiac lesions	Cardiac fibrinoid	Splenic fibrinoid	Liver thrombi and necrosis	Pulmonary thrombi and hemorrhage	Adrenal hemorrhage
		mg.	*hrs.*								
		4	18	3	3	1	3	2	1	3	0
			24	12	0	8	2	3	2	4	0
			6	8	1	4	2	7	1	8	0
Small	1:80		12	7	3	2	2	6	1	6	1
rabbits		8	18	22	14	15	14	15	6	14	0
			24	9	4	7	3	5	3	7	1
			30	3	0	1	0	0	0	1	0
	1:1000	8	Simultaneous	4	3	2	1	1	0	4	1
Large		16	Simultaneous	5	5	2	2	3	2	5	4
rabbits	1:1000	32	Simultaneous	5	5	0	1	2	0	5	2

Kidney. The renal lesion resembled closely that produced by two injections of endotoxin (the generalized Shwartzman reaction). It

consisted of the presence of homogeneous, eosinophilic, refractile material within the lumina of the glomerular capillaries (Fig. 1), which usually was associated with tubular necrosis and interstitial hemorrhage. The material had the tinctorial features of fibrinoid, being purple-red with the Schiff stain, homogeneous purple to orange-purple with phosphotungstic acid-hematoxylin, reddish blue with crystal violet, yellow with van Gieson's stain, and a homogeneous blue with toluidine blue. In some animals which died within a short period, the material appeared somewhat more basophilic with hematoxylin, and was not closely applied to the capillary basement membranes.

In 85 per cent of the rabbits in which this material was present in the capillaries, gross cortical necrosis was observed. In occasional animals, only isolated areas of cortical infarction were noted, while in others the major portion of the kidney was necrotic (Fig. 8). In the latter, it was sometimes possible to demonstrate thrombi in the larger renal vessels. Saline suspensions of fresh glomeruli were prepared in the manner previously described[2] and examined in polarized light. It was found that the capillary material was intensely birefringent and had properties similar to the material which occurs in the glomeruli after two intravenous injections of endotoxin.

Heart. The cardiac lesions consisted of areas of muscle necrosis and mononuclear cellular reaction, muscle calcification, intravalvular hemorrhages, and coronary thrombi. The latter often were associated with typical areas of ischemic necrosis. Edema and vacuolization of the intima and media of the coronary arteries, similar to the lesion observed after a single injection of endotoxin,[1] were noted occasionally in the animals given only Liquoid, and were consistently observed in those given endotoxin and Liquoid. Hemorrhages in the myocardium were of such common occurrence, especially in those animals given large amounts of Liquoid, that they were not tabulated.

Fibrinoid also was found in the hearts of the animals given either Liquoid alone or endotoxin and Liquoid. In the animals given only Liquoid, however, it lay free in the pericardium or beneath the atrial endocardium, while in the animals given endotoxin in conjunction with Liquoid it appeared as a layer of acidophilic, refractile material lying beneath the endothelium of the coronary arteries and within the heart valves (Figs. 2, 4, and 5). The incidence of fibrinoid in the hearts of the animals given only Liquoid was low, comparable to that observed after a single intravenous injection of endotoxin.[1] When endotoxin was given in conjunction with Liquoid, the highest incidence

of cardiac fibrinoid occurred when the interval between injections was 18 hours (Table IV).

Spleen. The splenic lesion was similar to that observed after two injections of meningococcal toxin.[2] It consisted of the accumulation of material within the splenic sinusoids which, like the material in the glomerular capillaries, had the morphologic and tinctorial features of fibrinoid. Its accumulation was sufficient in many instances to produce areas of ischemic necrosis which had the gross appearance of infarcts.

Liver. The changes in the liver consisted of areas of hemorrhage and necrosis, associated with the presence of acidophilic, slightly granular, thrombotic material in the efferent veins (Fig. 3). Occasionally only perivascular hemorrhages were observed, but in 2 animals these were associated with the presence of fibrinoid beneath the endothelium of branches of the hepatic artery. In many instances, especially in those animals in which there was a short interval between the injections of toxin and Liquoid, no thrombi were observed, but the sinusoids contained material which resembled that seen in the splenic sinusoids and renal glomerular capillaries.

Lungs. Pulmonary thrombi or emboli, edema, and hemorrhage were consistently noted in the animals given large amounts of Liquoid alone, or in those in which there was a short interval between the injections of toxin and Liquoid. The thrombi or emboli were for the most part confined to the capillaries, but in those given toxin in addition to Liquoid they were observed also in large branches of the pulmonary arteries (Fig. 9). The thrombi, or emboli, were often homogeneous and deeply eosinophilic, and were purple-red when stained by the periodic acid-Schiff method.

Adrenal Gland. Adrenal hemorrhage was noted in only a small percentage of the animals. It occurred with equal ffequency in animals given either Liquoid or endotoxin and Liquoid, except in the group of large rabbits. In these animals it was observed more often in the group given simultaneous injections of toxin and Liquoid. Most cases showed diffuse hemorrhage which replaced varying portions of the gland, but in one animal there were multiple infarcts in both adrenal glands (Fig. 10).

Brain. Changes in the central nervous system were observed only in those animals given endotoxin in association with Liquoid. The cerebral lesion consisted of the presence of material within the capillaries of the choroid plexus which was strikingly similar to that observed in the renal glomerular capillaries. Occasionally it appeared to lie beneath the endothelium, but in most cases it lay free in the lumen. It

usually was not applied closely to the basement membranes and in this respect differed from the renal glomerular material (Fig. 6). Its staining reactions, however, were similar to those of the material in the kidneys and spleens. Thrombi within the cerebral vessels also were noted occasionally (Fig. 11), in one instance lying in close proximity to an area of acute encephalomalacia.

Ears. Lesions involving the ears were observed only in those animals given endotoxin in association with Liquoid, particularly in those in which the interval between endotoxin and Liquoid was 18 hours. These changes occurred approximately 6 to 8 hours after the injection of Liquoid and increased in severity until death or sacrifice. Involvement of the ears was almost invariably associated with the development of bilateral renal cortical necrosis.

Grossly, the ears became boggy, purple-blue, and drooped. Microscopically, many of the vessels were occluded by dense masses of homogeneous, eosinophilic material resembling fibrinoid which replaced the wall of the vessel and extended into the perivascular spaces. This was associated with marked hemorrhage, edema, and varying degrees of cellular infiltration (Fig 7).

Other Organs. Hemorrhage within the thymus was observed in animals given large amounts of Liquoid and in those given endotoxin in association with smaller amounts of Liquoid. In many instances the entire gland appeared as a hemorrhagic mass of tissue. Intestinal hemorrhage was observed in 3 animals, 2 of which had been given endotoxin and Liquoid. In those latter animals many of the vessels in the hemorrhagic area were occluded by thrombi, and fibrinoid was present beneath the endothelium of the vessels.

The Effects of Dextran Sulfate and PVAS

The lethal effect of dextran sulfate and PVAS was not as marked as that observed with comparable amounts of Liquoid. None of the animals given 50 mg. of PVAS, for example, died within 12 hours. When given in conjunction with endotoxin, however, small amounts of these materials showed a pronounced lethal effect. Fifty per cent of the animals given 8 mg. of PVAS 6 hours after endotoxin died within 12 hours, and 40 per cent of those given a simultaneous injection of 10 mg. of dextran sulfate and endotoxin died within the same period. Further observations concerning the lethal effect of these substances in varying amounts and with varying dilutions of endotoxin will be documented elsewhere.[18]

The structural changes observed in these animals were similar to those described for the previous groups. No adrenal lesions were noted

in any of the animals, however, and no renal or splenic lesions developed in the animals given PVAS or dextran sulfate alone. The latter two lesions, however, developed with regularity when these polymers were given simultaneously with endotoxin (Tables V and VI). The material which accumulated in the renal glomerular capillaries and in the splenic sinusoids was similar to that observed in the Liquoid group, and had similar staining properties.

TABLE V

Structural Changes Following the Injection of PVAS Alone and in Conjunction with Meningococcal Toxin *

Dilution of toxin	Interval between injections	Number of rabbits	Dose of PVAS	Renal lesion	Cardiac lesions	Cardiac fibrinoid	Splenic fibrinoid	Hepatic thrombi and necrosis	Pulmonary thrombi and hemorrhage
	hrs.		*mg.*						
		4	8	0	2	0	0	0	2
None		4	16	0	3	1	0	1	4
		4	32	0	2	0	0	0	3
		4	50	0	2	0	0	3	2
1:1000	Simultaneous	4	8	3	2	1	3	1	4
1:80	6	4	8	2	3	1	4	1	2
	18	4	8	0	4	2	2	1	3

* All animals in these groups weighed 1 to 1.5 kg.

TABLE VI

Structural Changes Following the Injection of Dextran Sulfate Alone and in Conjunction with Meningococcal Toxin *

Dilution of toxin	Interval between injections	Number of rabbits	Dose of dextran sulfate	Renal lesions	Cardiac lesions	Cardiac fibrinoid	Splenic fibrinoid	Hepatic thrombi and necrosis	Pulmonary thrombi and hemorrhage
			mg.						
None		4	10	0	2	0	0	0	0
1:500	Simultaneous	10	10	9	5	0	6	5	5

* All animals in these groups weighed 1 to 1.5 kg.

DISCUSSION

The most characteristic gross lesion of the generalized Shwartzman phenomenon is bilateral renal cortical necrosis. This reaction occurs after two appropriately spaced intravenous injections of meningococcal endotoxin and does not take place, except under unusual circumstances,

after a single injection.[2,14] This lesion appears to develop as the result of occlusion of the glomerular capillaries by homogeneous material with the tinctorial features of fibrinoid.

In the present study it has been shown that three acidic polymers of large molecular size, possessing anticoagulant properties somewhat similar to heparin, are capable of producing an equivalent renal lesion when administered intravenously to rabbits previously or simultaneously given a single intravenous injection of endotoxin. In the case of Liquoid, the polymer alone can produce the renal lesion when given in sufficiently large amounts. Walton[4] has shown that dextran sulfate alone can produce a similar renal lesion when given in doses of 100 mg. per kg. With the material used in the present study, no renal changes occurred with doses as high as 50 mg. per kg., but larger amounts could not be tested because of the immediate lethal effect.[13] PVAS, administered alone in doses as high as 50 mg., produced no renal lesions.

Other lesions produced by two injections of endotoxin, discussed in the previous paper,[2] include accumulations of fibrinoid in the vessels and valves of the heart and the splenic sinusoids, and pulmonary and hepatic thrombi with areas of hemorrhage and necrosis. Entirely similar lesions were produced by the acidic polymers in animals given one injection of endotoxin and, in the case of Liquoid, by the polymer alone when given in sufficient dosages. It should be noted, however, that the incidence of fibrinoid in the coronary arteries and heart valves was considerably higher in the animals given endotoxin and Liquoid than with Liquoid alone, even when large amounts of the latter were given. It is of interest, also, that in the animals given endotoxin and Liquoid, the greatest incidence of cardiac fibrinoid occurred when there was an interval of 18 hours between the injections. Similarly, the splenic lesion occurred more frequently in those animals given both endotoxin and Liquoid than in those given Liquoid alone.

These differences in the incidence of various lesions emphasize the importance of endotoxin in "preparing" the animals. Its importance is shown also by the occurrence of cerebral and aural lesions in animals given endotoxin in addition to Liquoid. These observations suggest that certain changes occur following the administration of endotoxin which bring about localization of the fibrinoid in certain sites, and that an optimal interval exists for the administration of these polymers after an injection of toxin. The structural changes which occur after a single intravenous injection of endotoxin[1] may be important factors in the localization of this material.

In addition to structural changes, the possibility that endotoxin may in some way bring about further polymerization of the injected compound or that it may act to alter a compound in the circulating blood which then combines with the injected polymer should be considered also. The lethal effects of Liquoid, PVAS, and dextran sulfate are enhanced by a previous or simultaneous injection of endotoxin. Patat and Vogler[11] reported an increase in the toxicity of PVAS with increasing degrees of polymerization, and a similar observation was made by Walton[4] concerning the dextran sulfates.

Experimental observations, reported elsewhere,[16] have shown that a heparin-precipitable protein, apparently related to fibrinogen, appears in the blood of rabbits within 1 hour after an injection of endotoxin. Moreover, an injection of Liquoid at that time is followed by a precipitous drop in the amount of this heparin-precipitable protein, accompanied by an equally abrupt fall in the level of total blood fibrinogen. In this regard, the demonstration by Walton[4] of the *in vitro* precipitation of fibrinogen by dextran sulfate is of special interest. These observations suggest that fibrinogen may be altered by endotoxin, and that this alteration may constitute the most important "preparing" function of endotoxin.

The basic similarity between the changes observed in rabbits given two appropriately spaced injections of endotoxin and in those given endotoxin in association with Liquoid, PVAS, or dextran sulfate suggests that similar mechanisms are involved in the production of the lesions. The diffuse nature of the tissue lesions and the histologic appearance of the material occluding the blood vessels suggest some change in the circulating blood leading to vascular occlusion in various organs with subsequent ischemic necrosis. The demonstration that the lesions can be prevented by heparin[13] is further evidence in favor of the hypothesis that the reaction represents primarily a disturbance in the circulating blood. These features, and the experimental observations concerning fibrinogen, support the hypothesis that fibrinoid is derived from the circulating blood and that fibrinogen may play an important rôle in its formation.

Summary and Conclusions

Hybrid albino rabbits of mixed sexes were given an intravenous injection of sodium polyanetholsulfonate (Liquoid), dextran sulfate, or the sodium salt of polyvinyl alcohol polysulfonic acid ester (PVAS) in varying amounts, alone and in conjunction with an intravenous injection of meningococcal toxin.

A pronounced lethal effect was noted when Liquoid was given alone in large doses, or when given in small doses in association with meningococcal toxin. Comparable amounts of PVAS or dextran sulfate were relatively non-toxic when given alone, but when administered in association with endotoxin caused death in a majority of the animals.

Structural changes in the kidneys, heart, spleen, liver, adrenal glands, lungs, brain, and ears were noted in the animals given the acidic polymers in conjunction with endotoxin. The important histologic feature of the lesions was the presence of fibrinoid. This material was observed in the glomerular capillaries, pericardium, coronary arteries, heart valves, the splenic sinusoids, the capillaries of the choroid plexus, and in the vessels and perivascular spaces of the ears. In 85 per cent of the cases in which this material was present in the renal glomerular capillaries, gross cortical necrosis similar to that occurring in the generalized Shwartzman phenomenon was present. An interval of 18 hours between the injection of endotoxin and Liquoid appeared to be the optimal time for the development of cardiac fibrinoid lesions.

Hepatic thrombi and necrosis were observed with equal frequency in animals given large amounts of Liquoid and in those given endotoxin and smaller amounts of Liquoid. Bilateral adrenal hemorrhage and necrosis were observed in a small percentage of animals. Pulmonary hemorrhages and thrombi or emboli were an almost constant finding in the animals given large doses of Liquoid, and in those given a simultaneous injection of endotoxin and small amounts of Liquoid. Splenic and hepatic lesions were not produced by dextran sulfate or PVAS alone, but occurred with regularity when these were given in conjunction with endotoxin.

There is a basic similarity between the lesions observed in these animals and in those given two appropriately spaced intravenous injections of endotoxin (the generalized Shwartzman phenomenon). The morphologic features of the lesions, the widespread occurrence of fibrinoid, and the biologic activities of these acidic polymers with respect to fibrinogen support the hypothesis that fibrinoid is derived from the circulating blood and suggest that fibrinogen may be directly involved in its formation.

The photographs were made by Mr. Henry Morris.

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[*Illustrations follow*]

LEGENDS FOR FIGURES

All sections were stained with hematoxylin and eosin.

FIG. 1. Renal glomerulus showing accumulation of homogeneous, eosinophilic material in the capillary loops. From a rabbit given 8 mg. of Liquoid 18 hours after an intravenous injection of meningococcal toxin and killed 24 hours later. × 450.

FIG. 2. Coronary artery with segmental deposition of fibrinoid in the media. Of note is the abrupt loss of nuclei. Rabbit given 4 mg. of Liquoid 24 hours after an injection of meningococcal toxin and killed in 24 hours. × 450.

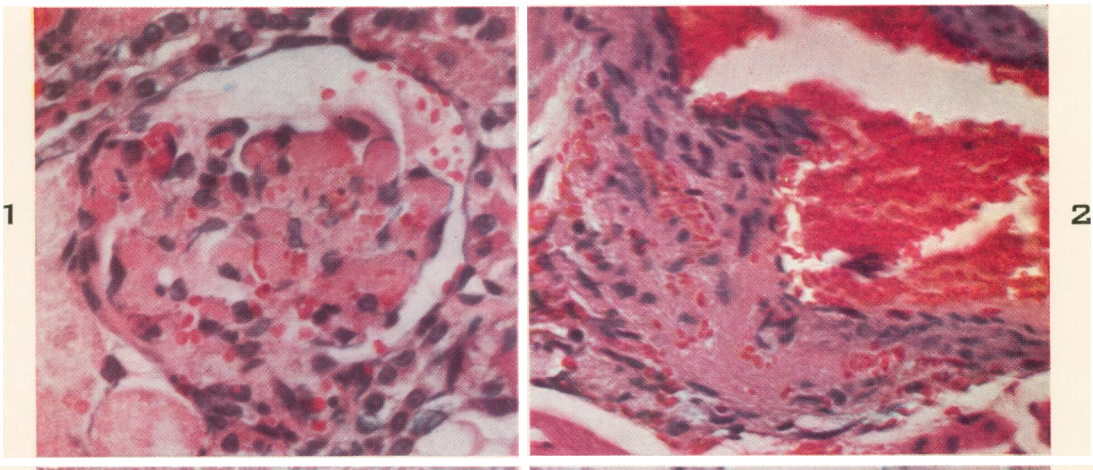
FIG. 3. Section from the liver of rabbit given 8 mg. of Liquoid 24 hours after an injection of meningococcal toxin and killed in 24 hours. Of note is an extensive area of hemorrhage and necrosis. × 100.

FIG. 4. Coronary artery showing extensive subendothelial deposition of fibrinoid. Rabbit given 4 mg. of Liquoid 18 hours after an injection of meningococcal toxin; killed in 30 hours. × 450.

FIG. 5. Cardiac valve leaflet showing extensive hemorrhage and a minimal deposition of fibrinoid. Rabbit given 8 mg. of Liquoid 12 hours after an injection of meningococcal toxin. Killed in 30 hours. × 100.

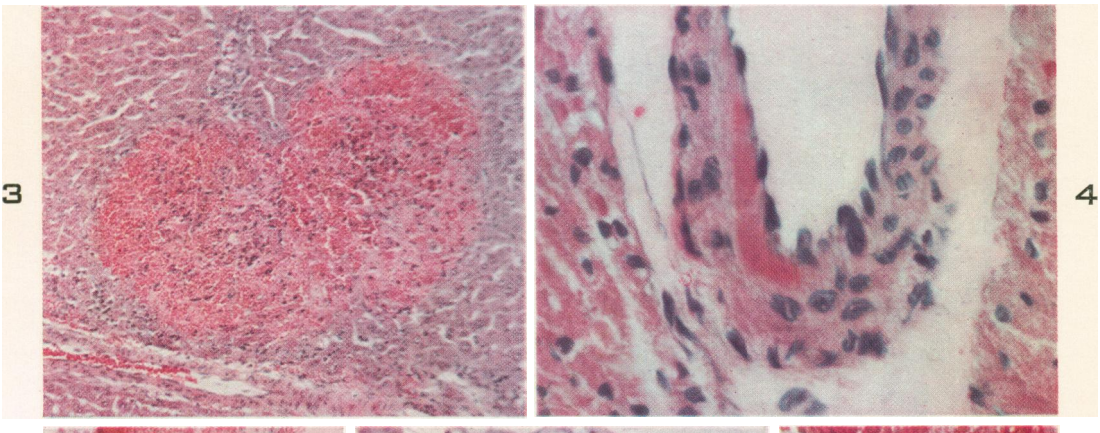
FIG. 6. Section from the choroid plexus of a rabbit given 8 mg. of Liquoid 18 hours after an injection of meningococcal toxin. The accumulation of material in these capillaries is similar to that in the glomerular capillaries. In the lower vessel it appears to be beneath the endothelium but otherwise it is intraluminal. Animal killed 30 hours after the injection of Liquoid. × 250.

FIG. 7. Section from the ear of a rabbit given 8 mg. of Liquoid 18 hours after an injection of meningococcal toxin and killed 30 hours later. There is extensive hemorrhage and most of the media of the artery is replaced by fibrinoid. Similar material is present also in the lumen of the vessel. × 250.



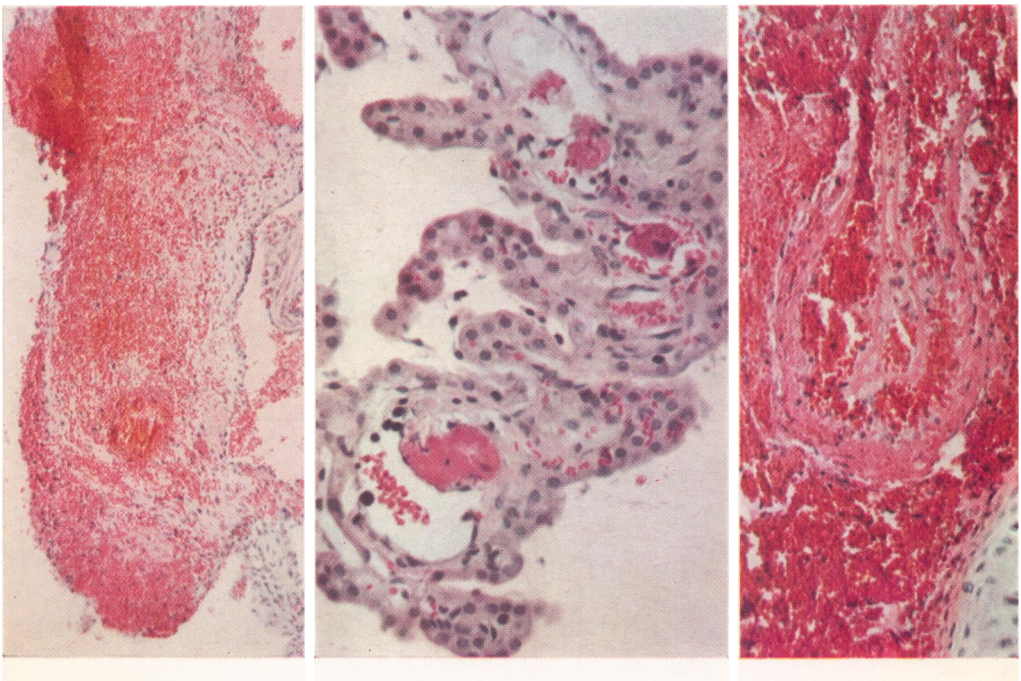
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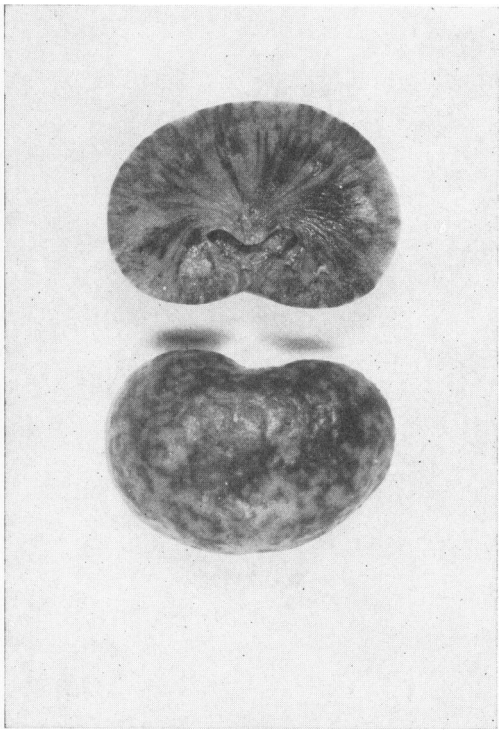
FIG. 8. Gross appearance of the kidney from a rabbit given a simultaneous intravenous injection of meningococcal toxin (dilution 1:1000) and 8 mg. of Liquoid. The animal died within 24 hours. Extensive hemorrhage and necrosis involve both cortex and medulla. Formalin fixation.

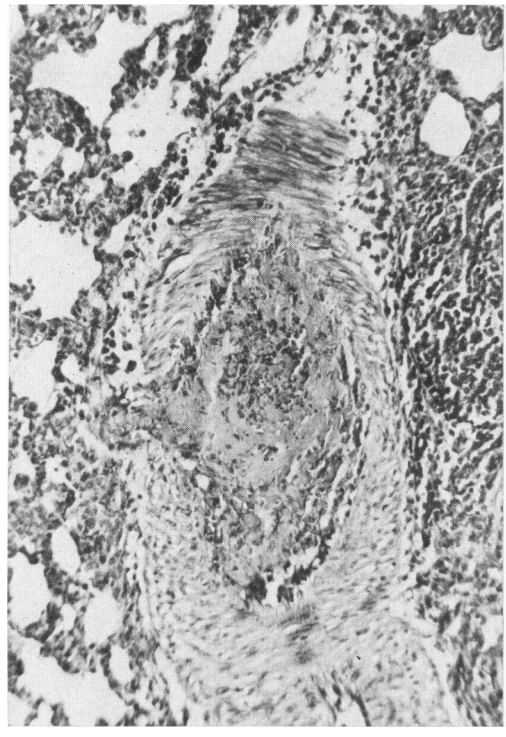
FIG. 9. Section from the lung of rabbit given 8 mg. of Liquoid 18 hours after an intravenous injection of meningococcal toxin and killed 30 hours later. There is a thrombus or embolus in a large branch of the pulmonary artery. × 150.

FIG. 10. Section from the adrenal gland of a mature rabbit given 32 mg. of Liquoid and killed in 24 hours. Of note is an extensive area of infarction. × 65.

FIG. 11. Cerebral vessel showing the presence of a thrombus with numbers of enmeshed cells and nuclear débris. Rabbit given 8 mg. of Liquoid 18 hours after an injection of meningococcal toxin; killed in 30 hours. × 250.

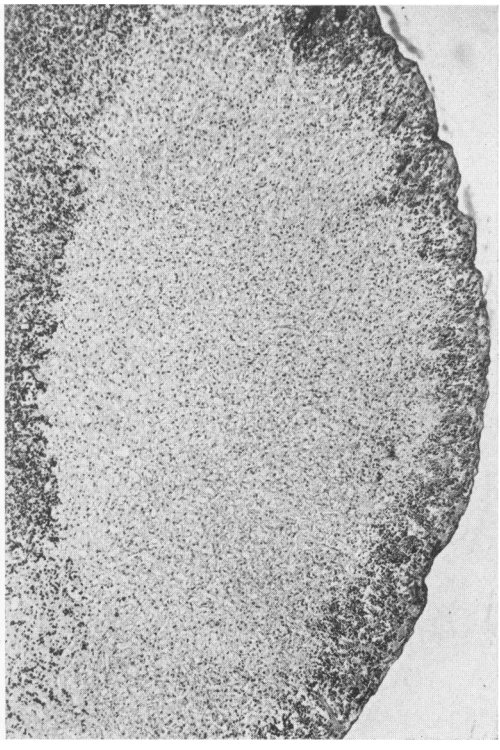
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